High-Force Linear Iron-core Fine-tooth Motor

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***Abstract*—This paper presents the design and testing of a high force linear *fine-tooth* iron-core permanent magnet motor. We present the design and experimental results in comparison with a commercially-available iron-core linear motor. Our new fine-tooth motor shows predicted shear stress improvements of 28 % at the prototype practical power level of 10 W/mm and 85 % at an anticipated ultimate root mean square (RMS) current density limit in the coil wires of 50 A/mm^2, relative to the conventional iron-core motor. This force performance of our new motor is also experimentally validated. The new motor shows strong potential for industry applications requiring precision motion along with high speed of production, such as in photo-lithography machines.**

***Index Terms*—High force, iron-core permanent magnet motor, linear motor, multi-phase motor**

I. Introduction

In order to meet the increasing industrial demand for high acceleration and high accuracy to achieve high throughput, such as in semiconductor lithography, a new fine-tooth linear motor is designed, prototyped, and compared to a conventional linear iron-core motor in both simulations and experiments. The new motor has the magnetic design features of fine teeth, narrow slots with high slot aspect ratio, five phases, and a moving Halbach magnet array, so as to generate high shear stress density with low force fluctuation.

There have been a number of studies on high torque/force motors over many years. In terms of rotary machines, a low-mass high-torque low-loss permanent magnet (PM) synchronous motor is presented in [1], [2] for the MIT Cheetah robot or for a quadruped robot in general targeting high speed motion. They describe the optimized rotary motor design producing a significant torque density (Nm/kg) increase compared to a commercial off-the-shelf motor of similar size. A commutation algorithm, called fine grain commutation, is introduced in [3] where the slot currents are individually controlled to ideally utilize all harmonics of stator and rotor magneto-motive forces (MMFs) to maximize the torque performance.

As for linear high-force motors, various tubular linear motors have been studied over many decades. Iron-cored tubular PM linear motors are introduced in [4]–[7], a surface wound tubular PM linear motor is discussed in [8] with the configuration of moving magnets and stationary coils, an ironless tubular PM linear motor is studied in [9] with long moving coils and short stationary Halbach magnet array, and tubular linear induction motors are introduced in [10], [11]. In addition, the design and testing of a high-acceleration moving permanent magnet linear motor in a double-sided configuration is presented in [12]. By reducing the mass of the moving magnet track without back iron, this motor design shows a high force density and acceleration. However, due to the magnetic configuration of 4 coils/poles and 2 magnets with 2-phase commutation, a significant force fluctuation is observed. This high force ripple issue is partially resolved in [13], [14] by modifying the magnetic configuration to 4 coils/poles and 3 magnets with 4-phase current commutation, which is a counterpart of the conventional 3-4 combination (3-phase-4-magnet) motor discussed in Section II.

Although a variety of motor research works have been conducted to achieve high torque/force over decades, it is difficult, especially in linear machines, to generate high force while simultaneously achieving low force fluctuation. In this paper, we present the magnetic design of a new linear iron-core PM motor which can simultaneously provide high thrust and low force fluctuation. The magnetic design approach and design parameter selection process of our new motor is discussed in Section II. We then introduce our experimental testbed and present force performance comparison between our new motor and a conventional iron-core PM motor in Sections III and IV, respectively, followed by conclusions in Section V.

II. New Fine-tooth Motor Design

In this section, we discuss the design of our new linear iron-core motor and compare its expected performance with a conventional motor. Fig. 1a shows the schematic magnetic design of the conventional linear iron-core motor (Tecnotion TL18). There are three phase windings wound on three iron-core teeth, magnetically interacting with four magnets in the magnet track. This is why we often call this type of motor a 3-4 combination motor, where 3 teeth/coils and 4 magnets compose a fundamental magnetic unit to generate thrust. Note that we define the length of this basic motor unit as λ_u as shown in the figure.

Our new motor design, on the other hand, has multiple fine teeth closely packed together as illustrated in Fig. 1b. Such a stator armature design is advantageous in that 1) we can achieve a stator MMF close to an ideal sinusoidal waveform to reduce force ripple or to an optimal triangular waveform to increase the thrust [3], 2) the stator material is better used

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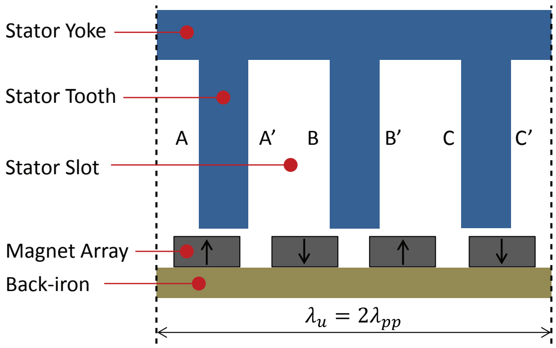
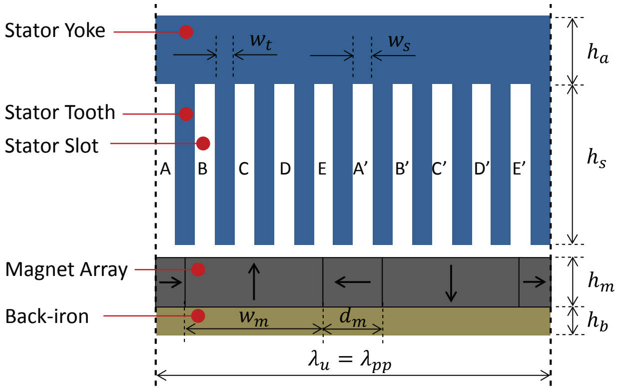


(a) Conventional 3-4 combination motor.



(b) New fine-tooth motor.

Fig. 1. Schematic design comparison of two linear iron-core motors over one fundamental motor unit of λ_u. Note λ_{pp} is the magnet pole-pair pitch (e.g. from N to N).

as the magnetic flux is more uniformly distributed, and 3) the cogging and the force ripple can be significantly reduced by a small angle of magnet skew, due to the small tooth pitch. Note that the schematic in the figure shows a 5-phase stator, but it can be any number of phases. Since we have fine teeth in our new motor design, we call our motor as a *fine-tooth* motor. For a magnet track, we use the Halbach array pattern to further increase the thrust performance. Another key advantage in our new magnetic design of fine teeth and Halbach magnet array is that we can achieve a significant reduction on the vibrational and acoustic noise of linear iron-core motors. This motor noise issue is discussed in detail in [15].

The key geometric design parameters of our new motor are indicated in Fig. 1b, and Table I shows their descriptions. For the selection of each design parameter, we change only that parameter value while keeping the other the same and select the value that shows the maximum force performance. Fig. 2 compares the shear stress of the new motor design variants of the magnet thickness. The shear stress, τ is obtained by

$$\tau = \frac{F_{thrust}}{A} \quad (\mathrm{N/mm^2}) \tag{1}$$

where F_{thrust} is the thrust force calculated by the finite element method, and A is the force-generating area (= magnet array length $\times$ motor depth) in the working air-gap of a motor. Note that the shear stress τ is plotted against the power

TABLE I
FINAL DESIGN PARAMETERS OF OUR NEW FINE-TOOTH MOTOR

Parameter	Description	Value
N_{slot}	Number of slots per pole (# of phase)	5
w_t, w_s	Tooth and slot width	2 mm
h_a	Thickness of armature yoke	15 mm
R_{slot}	Slot aspect ratio (h_s/w_s)	15
N	Number of winding turns per slot	126
AWG	Wire gauge for winding	23
f_{magnet}	Fraction of vertical magnets (w_m/λ_p)	0.7
PM_{thick}	Magnet aspect ratio (h_m/w_m)	0.5
h_b	Magnet back-iron thickness	4.76 mm
D	Motor depth	52 mm

dissipation per unit $P_{diss,unit}$, which is the instantaneous peak conductor power loss of

$$P_{diss,unit} = I_p^2\, R \quad (\mathrm{W}) \tag{2}$$

where I_p and R are the instantaneous peak current and coil resistance over a basic motor unit of λ_u.

Notice in Fig. 2 that regardless of the magnet thickness, our new fine-tooth motor shows higher shear stress than the conventional motor at all power levels. Note that our fine-tooth motor is expected to generate higher shear stress even with the same magnet array pattern and same thickness as the conventional motor (light blue line), as shown in the plot with the black line. The force performance is then enhanced further when a Halbach array pattern is used with thicker magnets. We observe in the figure that the fine-tooth motor designs with $PM_{thick} \geqslant 0.5$ (green, red, and blue curves) show significant shear stress increases compared to the conventional motor design (light blue curve). Thicker magnets, however, increase the moving mass and hence decrease the overall stage acceleration. We select the parameter value of $PM_{thick} = 0.5$ since it is expected to result in the maximum possible acceleration of our linear stage testbed. With the selected magnet thickness, it is expected to have 50 % shear stress increase (from 0.098 N/mm^2 to 0.147 N/mm^2), compared to the conventional motor

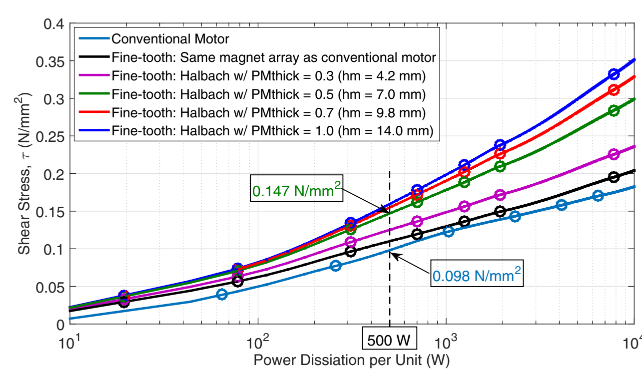


Fig. 2. Force capability comparison among a number of design variations to select the parameter, PM_{thick}.

at our testbed's practical power level of 500 W per motor unit as shown in Fig. 2.

The other design parameters are also determined by the same selection process, which is discussed in detail in [15]. Table I organizes the final values of the key design parameters of our new fine-tooth motor. The final magnetic design has five phases with the tooth/slot width of 2 mm, which results in the tooth pitch λ_t, the magnet pole pitch λ_p, and the fundamental unit length λ_u as

$$\begin{cases} \lambda_t = w_t + w_s = 4\,\text{mm} \\ \lambda_p = N_{slot}\,(w_t + w_s) = \lambda_u/2 = 20\,\text{mm} \\ \lambda_u = \lambda_{pp} = 2\lambda_p = 40\,\text{mm}. \end{cases} \tag{3}$$

The selected slot aspect ratio value of $R_{slot} = 15$ makes the slot depth h_s equal to 30 mm. For the magnet track, we have the magnet sizes as

$$\begin{cases} w_m = f_{magnet}\lambda_p = 14\,\text{mm} \\ d_m = (1 - f_{magnet})\lambda_p = 6\,\text{mm} \\ h_m = w_m \times PM_{thick} = 7\,\text{mm} \end{cases} \tag{4}$$

where w_m, d_m, and h_m are the width of vertical magnets, width of horizontal magnets, and magnet thickness, respectively, as illustrated in Fig. 1b.

With the final design parameter values, the expected shear stress is simulated and compared to the conventional motor as shown in Fig. 5. Note that we use in Fig. 5a the power dissipation per length for the abscissa. This is to take the motor unit length difference into account for a fair comparison. The shear stress is also plotted against the RMS current density in wire (current divided by a wire cross-section area) in Fig. 5b, since the RMS current density is directly related to the thermal limit of a motor. As can be seen from the shear stress comparison, our new fine-tooth motor (black curve) shows higher shear stress than the conventional motor (blue curve) at every power level. Specifically at a lower power level of 10 W/mm, our fine-tooth motor is expected to generate a 28 % higher shear stress of 0.115 N/mm^2 than the conventional motor's shear stress of 0.090 N/mm^2. We observe even higher shear stress increase of 85 % from 0.151 N/mm^2 to 0.280 N/mm^2 at an anticipated ultimate RMS current density level of 50 A/mm^2, showing significant force potential of our new fine-tooth motor.

Our new fine-tooth motor design is also advantageous in lowering the force ripple. Fig. 3 compares the simulated normal and tangential force ripples of the fine-tooth motor with the conventional motor. Both comparisons are made with skewed magnets at the motor unit power level of 500 W. As can be seen in the figure, our new motor generates smaller force ripple. Specifically, we have a significant peak-to-peak force ripple reduction with the ratio of 9-to-1 (35.8 N to 4.0 N) and 5-to-1 (20.5 N to 4.0 N) in the normal and tangential directions, respectively. It is further observed that our new motor contains almost solely fundamental force ripple while the conventional motor has a stronger fundamental with other higher harmonics of significant amplitudes. Note that this advantage of having less force harmonics can directly lead to less motor noise.

III. Experimental Setup

Our new fine-tooth motor is built based on the finalized design parameters shown in Table I, and the pictures of the stator armature and moving magnet tracks are presented in Fig. 4a and Fig. 4b, respectively. We have in the stator armature five-phase double-layered full-pitch concentrated windings with the coils of each phase connected in series. For the magnet tracks, we have both the non-skewed and skewed versions to experimentally observe the cogging reduction. Note that we choose the magnetic configuration of long stationary armature

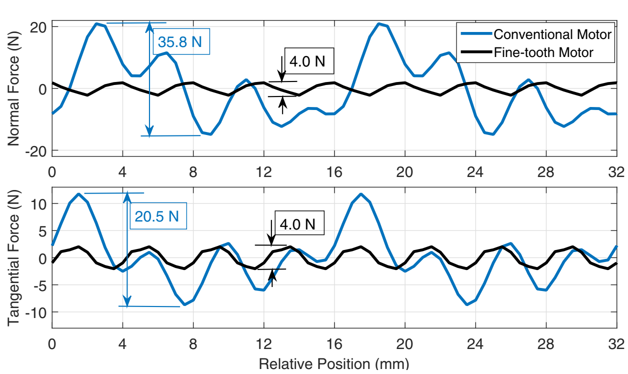
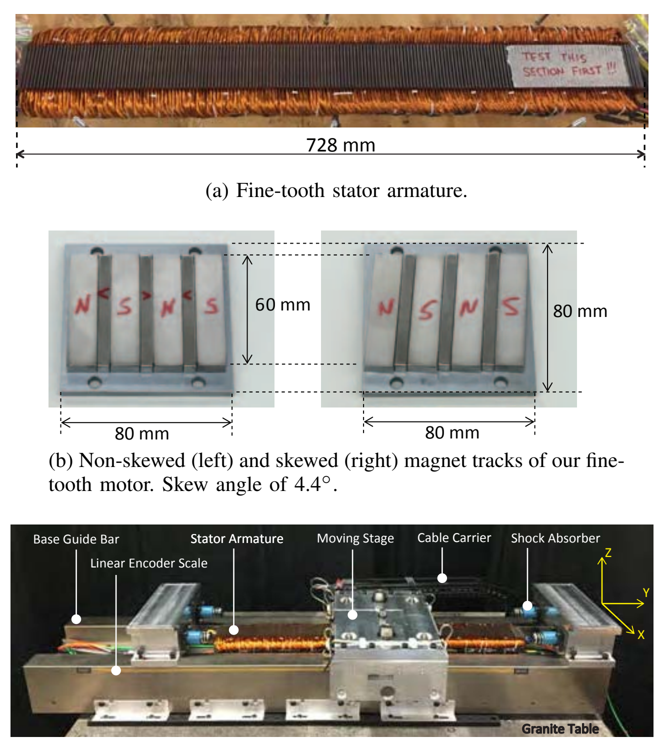


Fig. 3. Simulated force ripple comparison between the conventional motor and our new fine-tooth motor with final design parameters.



(a) Fine-tooth stator armature.

(b) Non-skewed (left) and skewed (right) magnet tracks of our fine-tooth motor. Skew angle of 4.4°.

(c) Completed linear stage testbed with our new fine-tooth motor.

Fig. 4. Pictures of experimental testbed and its linear motor components.

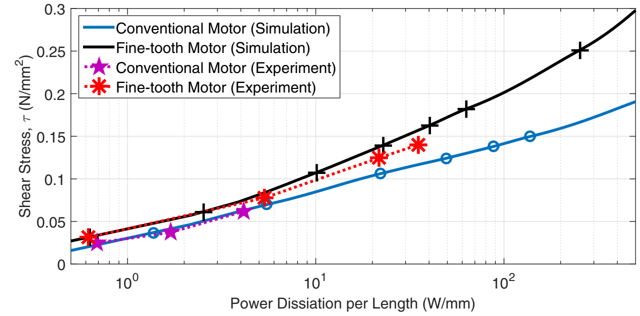
and short moving magnet because 1) it facilitates an stage acceleration increase due to the moving mass reduction and 2) the moving magnet configuration minimizes the number of umbilical cables that might cause assembly complexity and also undesired cable dynamics.

In order to validate the enhanced force performance of our new fine-tooth motor, compared to the conventional 3-4 combination motor, an experimental linear stage testbed is designed and built. Fig. 4c shows a picture of the stage setup on a granite table with our new fine-tooth motor implemented. The coordinates in the figure show the moving direction of Y, the cross-moving direction of X, and the normal direction of Z. Note that we use the same setup for the conventional motor test, but with only the stator and magnet track replaced. Our linear stage runs on air bearings for frictionless movement. Shock absorbers (SCS33-25 by ACE) are used at both travel ends to safely and passively stop the stage in an emergency runaway situation. The granite table and the base guide bar provide a mass ratio about 100-to-1 with respect to the moving stage mass of 11.5 kg, which is advantageous for handling the high reaction force at high acceleration operations. A dynamomter is configured by four triaxial load cells (Type 9250A4 by Kistler) under the stator to directly measure the motor forces (e.g. cogging). We use optical linear position sensors (read-head T1001 and scale RGSZ20 by Renishaw) to provide a real-time position feedback to the position controller with the interpolated resolutions of 0.1 μm. The stage is position-controlled at the -3dB-bandwidth of about 200 Hz. The position feedback control loop runs deterministically at 10 kHz in a real-time controller (PXI-8110 by National Instruments).

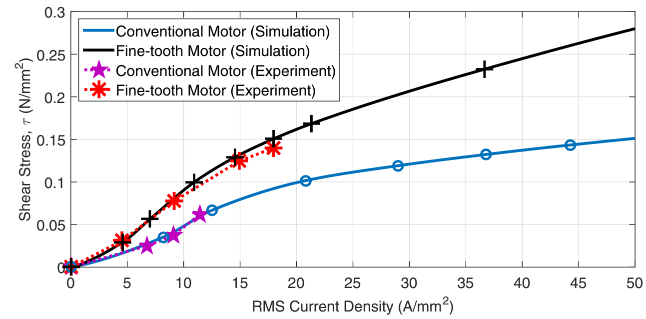
IV. Experimental Results

Our new fine-tooth motor is designed to provide higher force performance than the conventional 3-4 combination motor. In this section, we experimentally validate this by showing that the actual force performance of our new motor agrees with the expected one from the simulation discussed in Section II. The force performance of our new fine-tooth motor is presented in terms of shear stress as in (1), and is compared to the conventional iron-core motor both in simulations and experiments as shown in Fig. 5. The purple stars and the red asterisks are the experimental data points for the conventional and new motors, respectively. Note that skewed magnets are used for both motor types in simulations and experiments. We see in the plots that the data points of the conventional motor are limited to relatively low power (or current density) level because this motor emitted too much vibro-acoustic noise to allow experiments at higher power levels. The data points of our new motor, on the other hand, are limited by coil heating and travel length limits.

The experimental results of the conventional motor agree well with the finite element analysis while there is a bit of discrepancy with our fine-tooth motor as the power level increases. We speculate that this is likely due to the saturation limit differences between the finite element model and the



(a) Shear stress versus power dissipation per motor unit length.



(b) Shear stress versus RMS current density in coil wires.

Fig. 5. Simulated and experimental performance comparison between the conventional motor and our new fine-tooth motor.

actual core material. Despite this discrepancy, however, we can see that our motor behaves closely to the simulation results, showing higher force performance than the conventional motor. Specifically, at the power level of 20 W/mm, our new motor experimental data shows 19 % shear stress increase (from 0.104 N/mm^2 to 0.124 N/mm^2) than the simulated result of the conventional motor, and 57 % shear stress increase (from 0.079 N/mm^2 to 0.124 N/mm^2) at the RMS current density level of 15 A/mm^2. Given that the material saturation and the leakage issues can be mitigated by using cobalt-iron (e.g. Hyperco50) for the stator material rather than silicon-iron (M-19), we can expect more force potential with the material change, especially for a higher power range.

In our fine-tooth design, we can also achieve significant cogging reduction simply by skewing magnets. This is due to the facts that 1) unlike the conventional 3-4 combination motor, our new fine-tooth design allows solely the fundamental component of force fluctuation as shown in Fig. 3, 2) this fundamental force fluctuation can be significantly reduced by skewing magnets to span a full tooth pitch, and 3) the tooth pitch of our new motor is designed to be small enough, specifically 4 mm, for the magnets to be easily skewed without compromising the thrust. Fig. 6 shows the measured cogging forces of our fine-tooth motor with either the non-skewed or skewed magnets. Note that the skewing angle is calculated by

$$\theta_{skew} = atan\left(\frac{\lambda_t}{D}\right) = atan\left(\frac{4\text{ mm}}{52\text{ mm}}\right) \simeq 4.4^\circ \qquad (5)$$

so as to cover a tooth pitch λ_t with our motor depth D. As

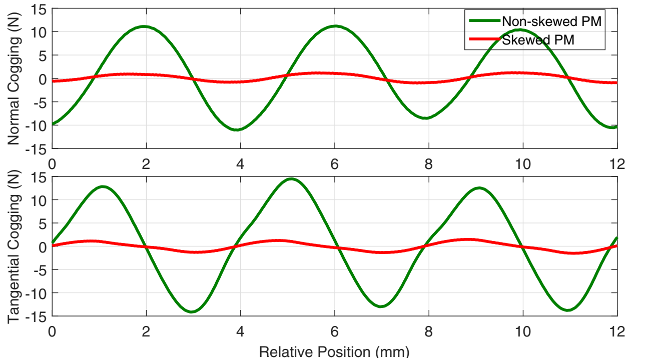


Fig. 6. Fine-tooth motor cogging force measurement comparison between non-skewed and skewed magnets. Mostly the fundamental component with the period equal to a tooth pitch, $\lambda_t = 4$ mm.

shown in the figure, skewing reduces the cogging significantly in both the normal and tangential directions. Specifically, the peak-to-peak amplitudes are reduced from 22.26 N (normal) and 28.62 N (tangential) to 2.22 N and 2.99 N, respectively. This is about a 10-to-1 (90 %) reduction in both directions. This significant cogging force reduction of our new motor directly leads to motor vibro-acoustic noise reduction, thereby helping to enhance system accuracy performance.

V. Conclusion

In this paper, we describe the design and testing of a new linear iron-core fine-tooth motor that can achieve high force and low force fluctuation at the same time. The enhanced force performance of our new fine-tooth motor are presented both in simulations and experiments, compared to a conventional 3-4 combination linear iron-core motor. With our new motor design, having fine teeth, narrow slots, multiple phase (5-phase), and a moving Halbach magnet array, we could achieve higher shear stress density than the conventional motor, specifically, predicted increase of 28 % at a power-per-motor-length level of 10 W/mm and even higher increase of 85 % at an RMS current density of 50 A/mm^2, achievable with a heavily-water-cooled system. The force fluctuation performance is also improved with our motor design. The force ripple at the power level of 500 W is reduced, compared to the conventional motor, by 89 % (9-to-1 reduction) and 80 % (5-to-1 reduction) in the normal and tangential directions, respectively. In addition, the cogging force is further decreased within our new motor design by slightly skewing magnets. We achieve a 90 % reduction (10-to-1) in both normal and tangential directions.

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